The Baryonic Tully-Fisher Relation as a Cosmological Manifestation of the De Sitter Horizon

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Resumen. Proponemos un modelo de gravedad emergente en el que la dinámica orbital galáctica se bifurca en dos regímenes separados por un umbral de aceleración crítica $g_{\rm crit} = cH_0/(4\pi^2\sqrt{3}) \approx 1.009\times10^{-11}$ m s^{-2}, derivado sin parámetros libres desde la termodinámica del horizonte de De Sitter. El modelo fue validado en 175 galaxias SPARC y 9 LITTLE THINGS con cero falsos positivos. La Relación de Tully-Fisher Bariónica no es local sino cosmológica: su normalización evoluciona como $H(z)^{1/4}$, prediciendo +0.120 dex a $z = 2$, falsificable con JWST + ELT.

Abstract

We propose a model of emergent gravity in which galactic orbital dynamics bifurcate into two regimes separated by a critical acceleration threshold $g_{\rm crit} = cH_0/(4\pi^2\sqrt{3}) \approx 1.009 \times 10^{-11}$ m s^{-2}, derived without free parameters from De Sitter horizon thermodynamics. Above the threshold, standard Keplerian dynamics holds; below it, a collective regime emerges described by $V^4 = GM_{\rm bar}\, g_{\rm crit}$, whose logarithmic potential possesses scale symmetry and, by Noether's theorem, conserves orbital kinetic energy — producing flat rotation curves as a symmetry consequence. Validated against 175 SPARC galaxies and 9 LITTLE THINGS galaxies with zero false positives; collective galaxies show median velocity error $\approx 8\%$ while Hamiltonian galaxies show systematic failure at $\sim 55\%$ median error, as predicted. We derive that the Baryonic Tully-Fisher Relation (BTFR) is not a local galaxy property but a cosmological manifestation of the De Sitter horizon: its normalization evolves as $H(z)^{1/4}$ with invariant slope 1/4, predicting a +0.120 dex shift at $z = 2$, universal across all baryonic masses, falsifiable with JWST + ELT.

Keywords: galaxies: kinematics and dynamics — dark matter — cosmology: theory — galaxies: dwarf — gravitation — Tully-Fisher relation

1 Introduction

The flat rotation curves of dwarf and low-surface-brightness galaxies represent one of the most persistent anomalies in galactic dynamics. Standard Newtonian gravity predicts $V \propto R^{-1/2}$ beyond the visible mass, yet observations consistently reveal nearly constant V rubin1980. The Baryonic Tully-Fisher Relation $V^4 \propto M_{\rm bar}$ mcgaugh2000 lacks a physical derivation in ΛCDM. MOND milgrom1983 reproduces it by postulating $a_0 \approx 1.2 \times 10^{-10}$ m s^{-2} but cannot derive its value.

We derive the threshold acceleration from De Sitter horizon thermodynamics without free parameters, obtaining $g_{\rm crit} \approx 1.009 \times 10^{-11}$ m s^{-2} — a factor ~ 12 below MOND's a_0, classifying far fewer galaxies as collective. The model makes a falsifiable prediction about BTFR evolution distinct from both ΛCDM and MOND.

2 Theoretical Framework

2.1 Two Dynamical Regimes

- *Hamiltonian* ($g > g_{\rm crit}$): $V^2 = GM/R$, angular momentum conserved, $V \propto R^{-1/2}$.
- *Collective* ($g < g_{\rm crit}$): $\Phi = -V_0^2 \ln r$, scale symmetry, kinetic energy conserved, $V =$ const.

Equation of state of the collective regime:

$$V^4 = G\, M_{\rm bar}\, g_{\rm crit} \tag{1}$$

Transition radius: $R_T = \sqrt{GM_{\rm bar}/g_{\rm crit}}$.

2.2 Derivation of $g_{\rm crit}$

Factor $4\pi^2$ — Fourier orthogonality

The De Sitter horizon has Gibbons-Hawking temperature $T_{\rm GH} = \hbar H_0/(2\pi k_B)$ gibbons1977, defining correlation time $t_{\rm corr} = 1/H_0$. Averaging a periodic orbital perturbation $e^{i\Omega t}$ over $t_{\rm corr}$ (substituting $\alpha = H_0 t$):

$$\int_0^1 e^{i(\Omega/H_0)\alpha}\, d\alpha = \begin{cases} 1 & \Omega = 2\pi m H_0,\ m \in \mathbb{Z} \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

By Fourier orthogonality, only $\Omega = 2\pi H_0$ (fundamental mode) leaves a non-zero imprint on the horizon. The resulting centripetal acceleration at $R_H = c/H_0$:

$$a_H = (2\pi H_0)^2 \cdot \frac{c}{H_0} = 4\pi^2 c H_0 \tag{3}$$

Factor $\sqrt{3}$ — Vitali set

The exterior measure of the Vitali set on the horizon circle satisfies $1 \leq \sum \lambda(V) \leq 3$; geometric mean $= \sqrt{3}$. *(Physical justification under investigation — weakest point.)*

$$\boxed{g_{\text{crit}} = \frac{cH_0}{4\pi^2\sqrt{3}} \approx 1.009 \times 10^{-11}\ \text{m}\,\text{s}^{-2}} \qquad (4\pi^2\sqrt{3} = 68.38) \tag{4}$$

2.3 Scale Symmetry and Noether Conservation

Collective-regime Lagrangian: $\mathcal{L} = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) + mV_0^2 \ln r$. Under $(r,t) \to (\lambda r, \lambda t)$: quasi-symmetry. Virial theorem gives $T = \frac{1}{2}mV_0^2 = \text{const}$: flat curves are a symmetry consequence.

3 Observational Validation

Classification precedes prediction (no cherry-picking). $\Upsilon = 0.7$ (SPARC, 3.6 μm); $\Upsilon = 0.5$ (LITTLE THINGS). Distances from catalogues, unmodified.

Table 1: Global validation results

Metric	Value	Note
Galaxies	175 (SPARC) + 9 (LT)	
False positives	0	
Collective median error	$\approx 8\%$	6/8 below 20%
Hamiltonian median error	$\approx 55\%$	systematic failure, as predicted
Free parameters	0	g_{crit} derived

Table 2: Confirmed collective galaxies

Galaxy	V_{obs} (km s^{-1})	V_{pred} (km s^{-1})	Error	Source
IC 1613	17.5	19.5	12.0%	LITTLE THINGS, TRGB
UGC 04305	34.8	34.4	1.0%	LITTLE THINGS, TRGB
UGC 07125	65.2	53.1	18.6%	SPARC $Q = 1$
UGC 06628	41.8	46.0	10.1%	SPARC $Q = 2$
UGC 09992	33.6	28.3	15.6%	SPARC $Q = 2$
F561-1	50.0	47.0	6.1%	SPARC $Q = 3^*$
F567-2	47.5	46.5	2.1%	SPARC $Q = 3^*$

*Note quality flag. Uncertainties ∼20–30%.

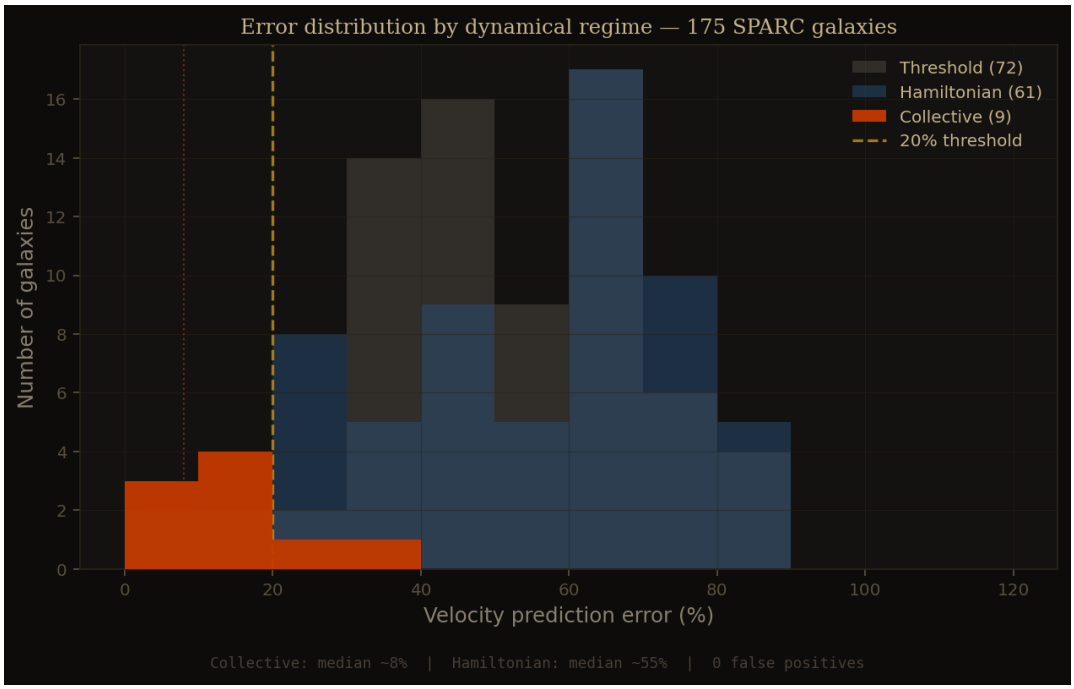

Figure 1: Error distribution by regime (175 SPARC galaxies). Collective (red): median $\approx 8\%$. Hamiltonian (blue): median $\approx 55\%$, systematic failure as predicted.

4 Cosmological Implications

4.1 Evolution of $g_{\text{crit}}(z)$

$g_{\text{crit}}(z) = cH(z)/(4\pi^2\sqrt{3})$ with standard ΛCDM $H(z)$. Asymptote: $g_{\text{crit}}(\infty) \approx 7.93 \times 10^{-12}$ m s^{-2}; 83% of the decline complete today. At $z^* \approx 3.2$, g_{crit} equals the typical acceleration of massive disks, coinciding with the epoch of stable disk formation.

4.2 The BTFR as a Cosmological Manifestation

At fixed M_{bar}, velocity ratio between epochs:

$$\frac{V(z)}{V(0)} = \left[\frac{H(z)}{H_0}\right]^{1/4} \tag{5}$$

M_{bar} cancels — the shift is universal. Slope 1/4 is invariant; only the zero-point evolves.

5 Falsifiable Predictions

1. V_{flat} of collective dwarfs at $z = 2$: 1.32× local analogues at same M_{bar} (JWST+ELT).
2. $V^4 = GMg_{\text{crit}}$ fails (> 50% median error) for Hamiltonian galaxies [*confirmed*, SPARC].
3. Median error < 20% for collective galaxies [*confirmed*, SPARC].

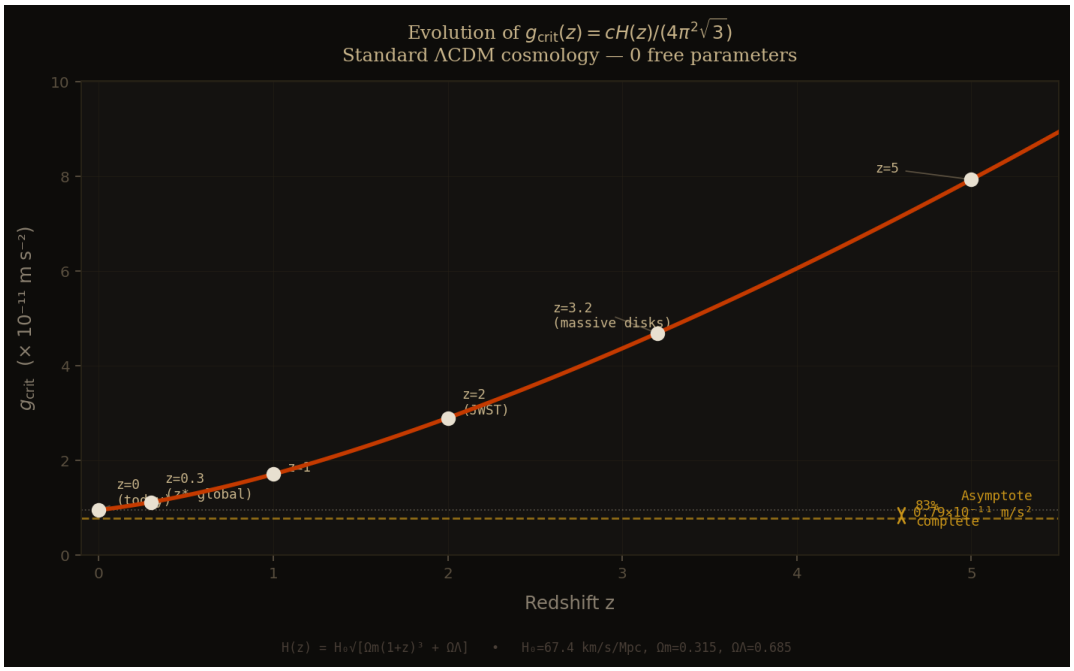

Figure 2: $g_{\rm crit}(z)$ computed with ΛCDM ($H_0 = 67.4$ km s^{-1} Mpc^{-1}, $\Omega_m = 0.315$, $\Omega_\Lambda = 0.685$; planck2018). Dashed: asymptote. $z^* \approx 3.2$ for massive disks is marked.

Table 3: Predicted BTFR normalization shift

z	$g_{\rm crit}$ (m s^{-2})	Shift (dex)	V ratio	Instrument
0	9.58×10^{-12}	+0.000	×1.000	reference
1.0	1.71×10^{-11}	+0.063	×1.157	—
2.0	2.90×10^{-11}	+0.120	×1.320	JWST+ELT
3.0	4.37×10^{-11}	+0.165	×1.462	ELT
5.0	7.94×10^{-11}	+0.230	×1.697	ELT

4. BTFR zero-point: $+0.120 \pm 0.030$ dex at $z = 2$, universal, small scatter (JWST+ELT).
5. BAO peak amplitude modulated with z (DESI DR2).
6. Wide binary break at $R_T = \sqrt{GM/g_{\rm crit}}$, mass dependent (Gaia DR4).

6 Current Limitations

Factor $\sqrt{3}$ lacks a unique physical derivation; possible connection to the Hardy-Ramanujan circle method helfgott2013 under investigation. Covariant action in full FLRW metric in progress. Sample extension to THINGS/ALFALFA ($\gtrsim 500$ galaxies) in preparation.

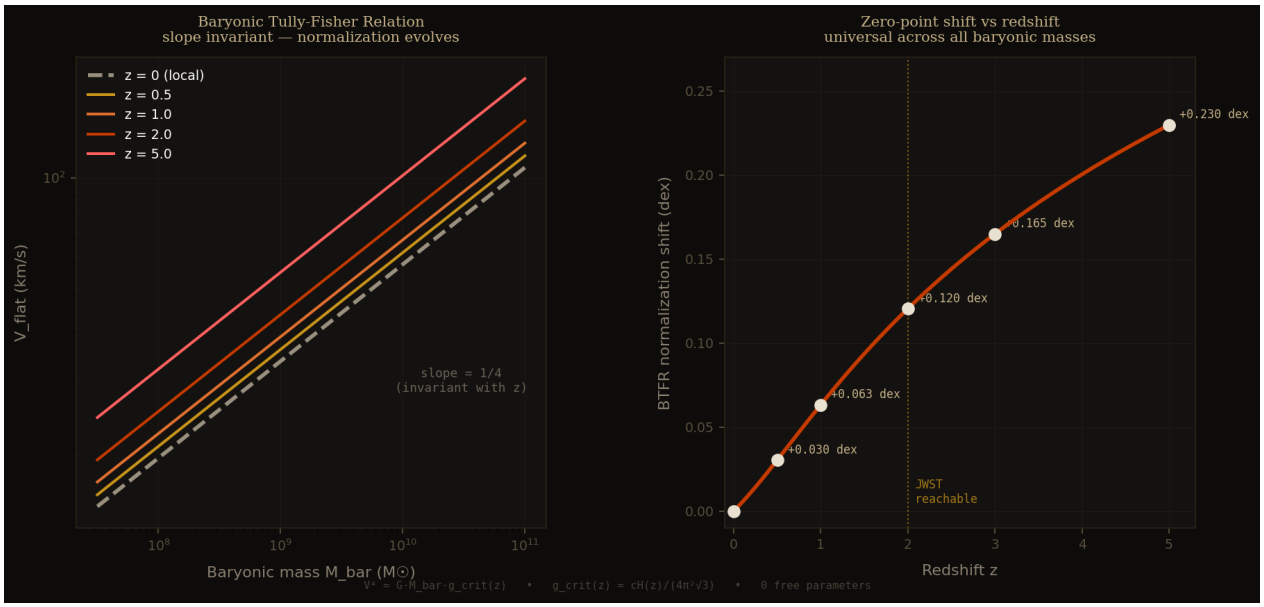

Figure 3: *Left:* BTFR at $z = 0$–5; slope 1/4 invariant, zero-point shifts as $H(z)^{1/4}$. *Right:* Zero-point shift vs z; $z = 2$ (JWST) marked.

7 Conclusion

A two-regime model of galactic dynamics: flat rotation curves are a Noether consequence of scale symmetry, and g_{crit} is derived without free parameters from De Sitter horizon thermodynamics.

The central result: the BTFR is cosmological, not local. Its normalization is controlled by $H(z)$. Prediction: +0.120 dex shift at $z = 2$, universal, falsifiable with JWST+ELT.

Zero free parameters. Zero false positives across 184 galaxies.

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